



Amino acids, Peptides & Proteins

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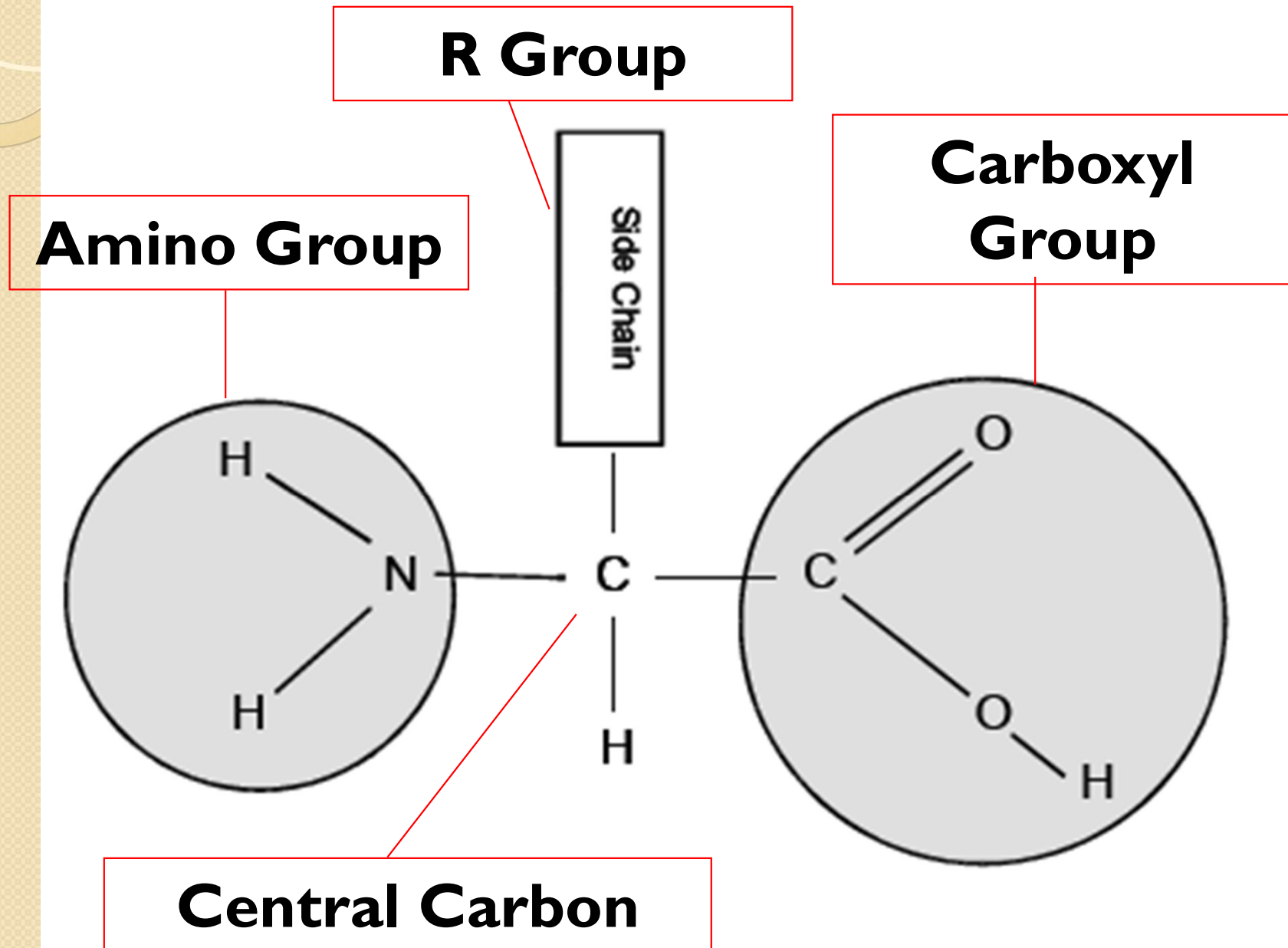
Chemistry for CSE

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Amino Acids: Building Blocks of Proteins

- Every **amino acid** has the same fundamental structure which consists of a central carbon bonded to an amino group, a **carboxyl group**, a hydrogen atom & a side chain (R).

An Amino Acid



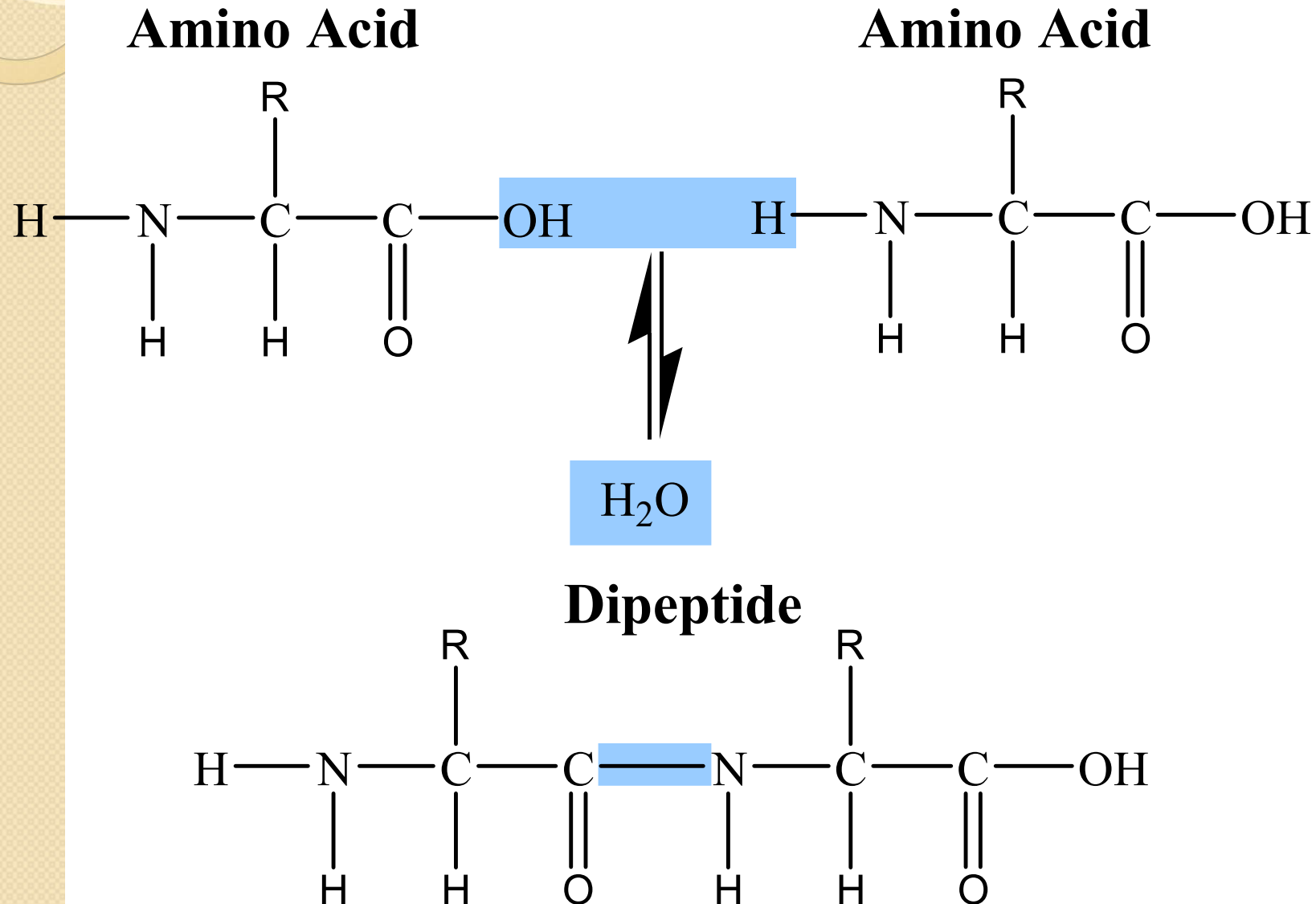
Amino Acids: Side Groups

- There are 20 different types of amino acids used in making proteins.
 - The essential structure is the same in all 20 molecules.
 - The 20 molecules differ in the side groups.

Amino Acids: Building Blocks of Proteins

- A **peptide bond** is a covalent bond formed.
- **Polypeptides** are polymers of amino acids linked together by peptide bonds, with the amino group of one acid joining the carboxyl groups of its neighbor.

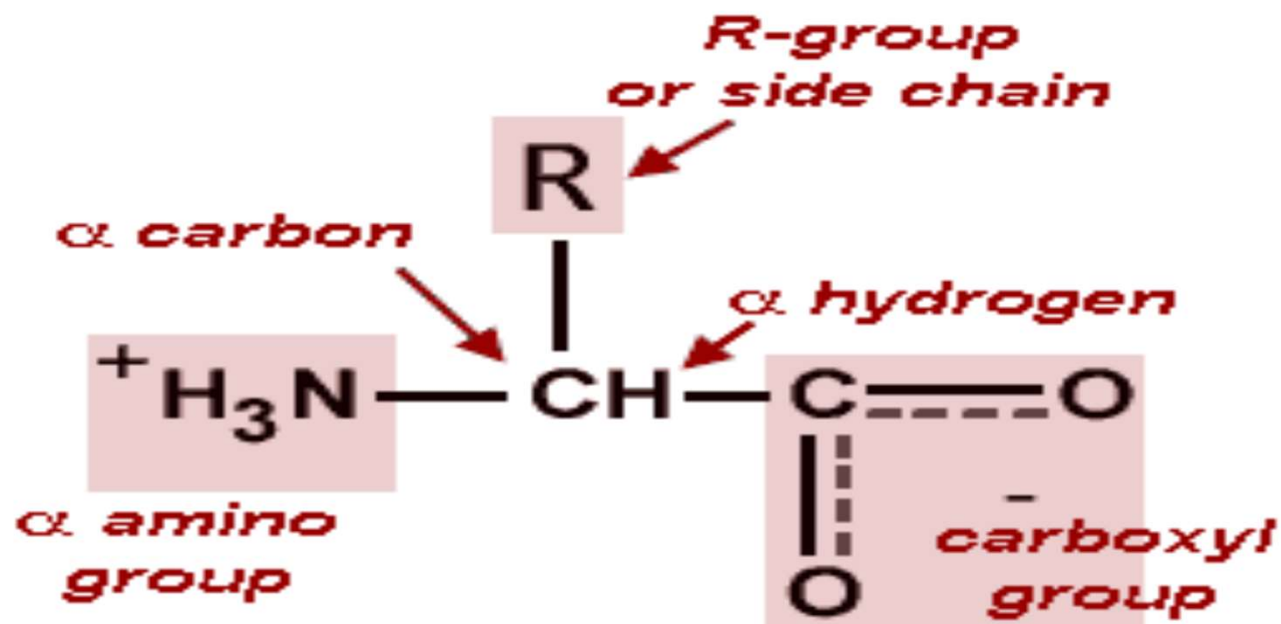
Amino Acids: Building Blocks of Proteins



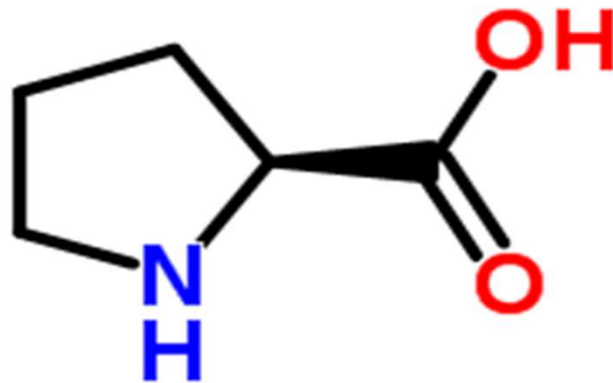
Amino Acids: Building Blocks of Proteins

- The peptide bond is formed when the OH group from the carboxyl group of one of the amino acids is removed and a hydrogen from the other amino acid is removed. The OH and H bond to form water.
- The formation of water from the two amino acids is called **dehydration synthesis**.
- The covalent linkage that is formed is known as a peptide bond, and the molecule that is formed by the linking of amino acids is called **polypeptide**.

At physiological pH (7.4), -COOH group is dissociated forming a negatively charged carboxylate ion (COO^-) and amino group is protonated forming positively charged ion (NH_3^+) forming Zwitter ion .



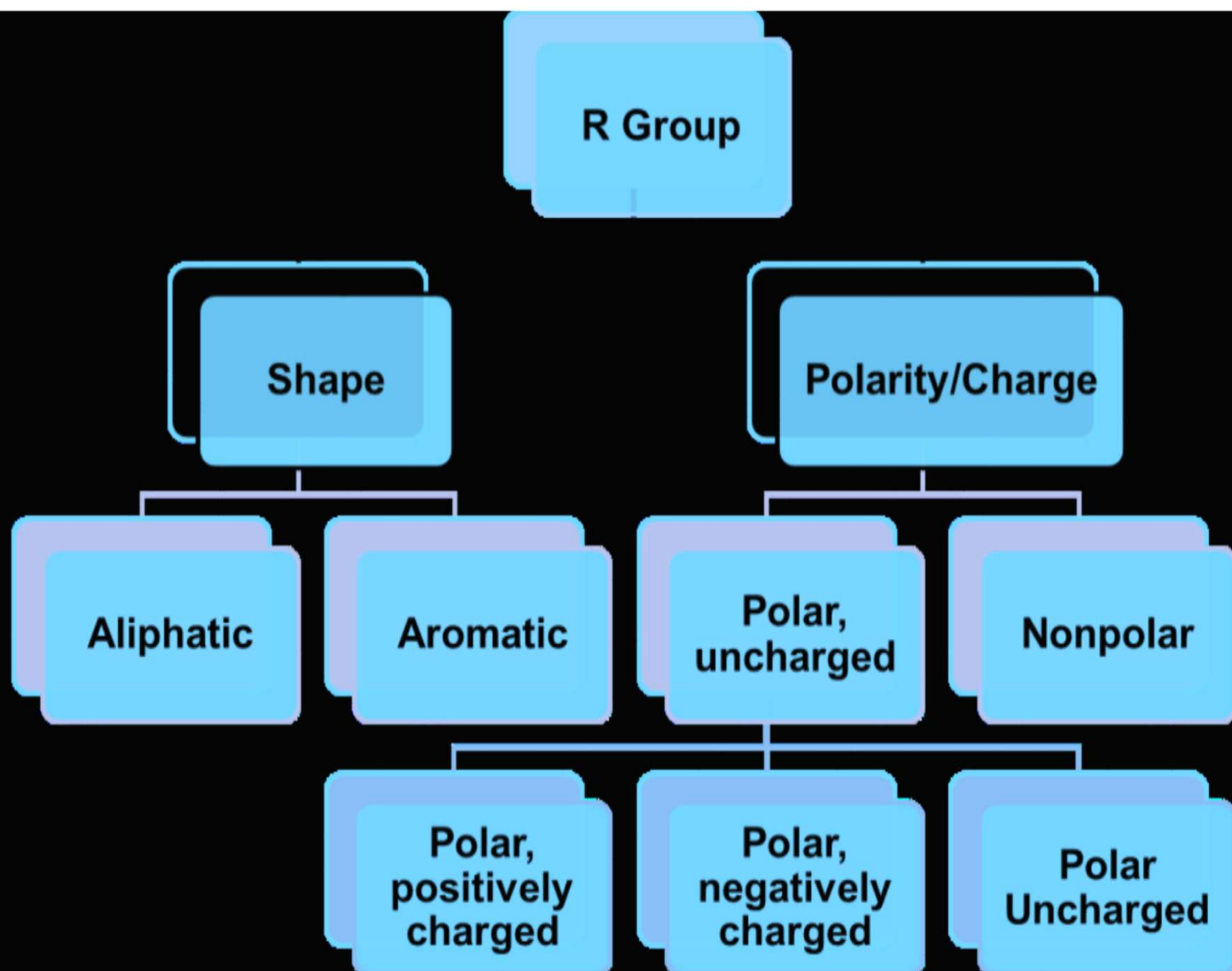
- Proline is an imino acid not amino acid .



Classification of Amino Acids:

- I. Classification by R group
- II. Chemical Classification
- III. Nutritional Classification
- IV. Metabolic Classification

Classification by R group



Each amino acid, aside from its name, has a three letter abbreviation and a one letter code.

Nomenclature

Amino Acid	3 letter code	1 letter code	Amino Acid	3 letter code	1 letter code
Glycine	Gly	G	Threonine	Thr	T
Alanine	Ala	A	Cysteine	Cys	C
Valine	Val	V	Tyrosine	Tyr	Y
Leucine	Leu	L	Asparagine	Asn	N
Isoleucine	Ile	I	Glutamine	Gln	Q
Methionine	Met	M	Aspartic Acid	Asp	D
Proline	Pro	P	Glutamic Acid	Glu	E
Phenyl alanine	Phe	F	Lysine	Lys	K
Tryptophan	Trp	W	Arginine	Arg	R
Serine	Ser	S	Histidine	His	H

Functions of Amino Acids

Apart from being the monomeric constituents of proteins and peptides, amino acids serve variety of functions.

(a) Some amino acids are converted to carbohydrates and are called as **glucogenic amino acids**.

(b) Specific amino acids give rise to specialised products, e.g.

- **Tyrosine** forms hormones such as **thyroid hormones**, (T3, T4), **epinephrine** and **norepinephrine** and a pigment called **melanin**.
- **Tryptophan** can synthesise a vitamin called **niacin**.
- **Glycine, arginine and methionine** synthesise **creatine**.

- Glycine and cysteine help in **synthesis of Bile salts.**

- Glutamate, cysteine and glycine synthesis **glutathione.**

- **Histidine** changes to **histamine** on decarboxylation.

- **Serotonin** is formed from tryptophan.

- Glycine is used for the synthesis of **haem.**

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- Pyrimidines and purines use several amino acids for their synthesis such as aspartate and glutamine for pyrimidines and glycine, aspartic acid, Glutamine and serine for purine synthesis

c) Some amino acids such as glycine and cysteine are used as detoxicants of specific substances.

(d) Methionine acts as “active” methionine (S-adenosylmethionine) and transfers methyl group to various substances by transmethylation.

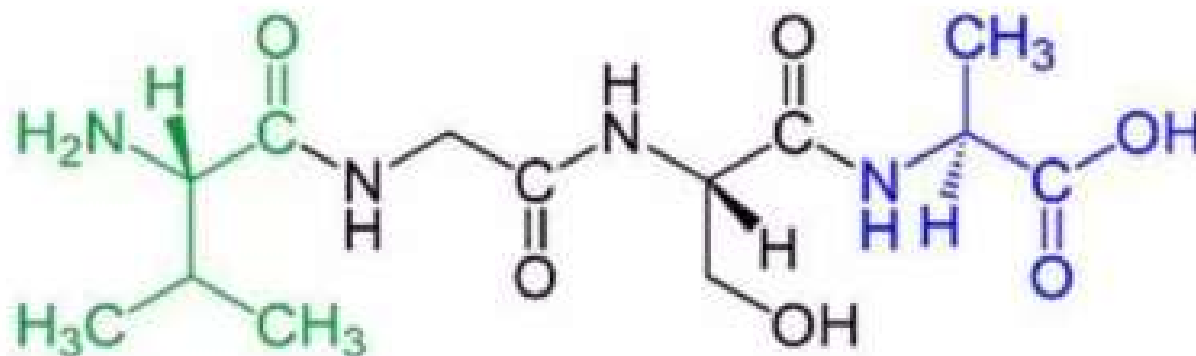
(e) Cystine and methionine are sources of sulphur



Peptides & Proteins

1.1 Peptide

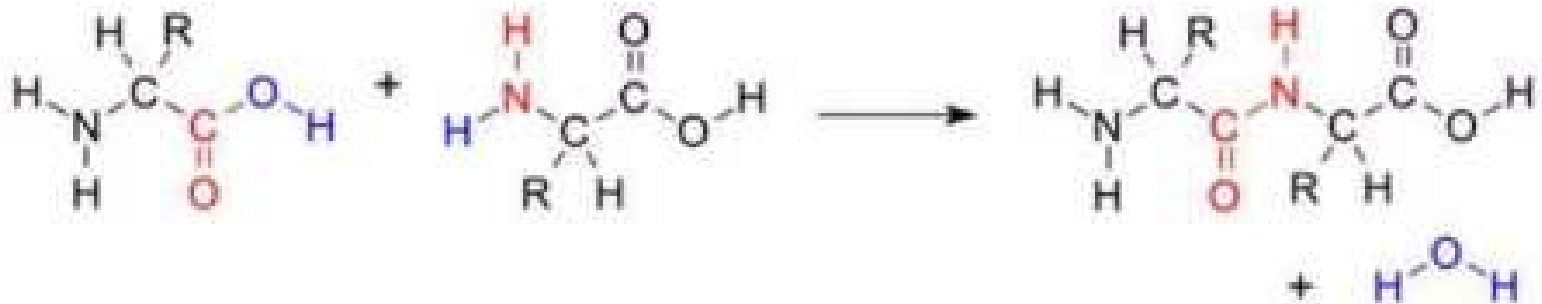
Peptides are short chains of amino acid monomers linked by peptide bonds which can be natural biological or artificially manufactured.



This is a tetrapeptide which includes Amino Acides:-
Valine, Glycine, Serine and Alanine

1.2 Peptide Bond

A **peptide bond** (also known as amide bond) is a covalent chemical bond linking two consecutive **amino acid** monomers along a peptide or **protein** chain.

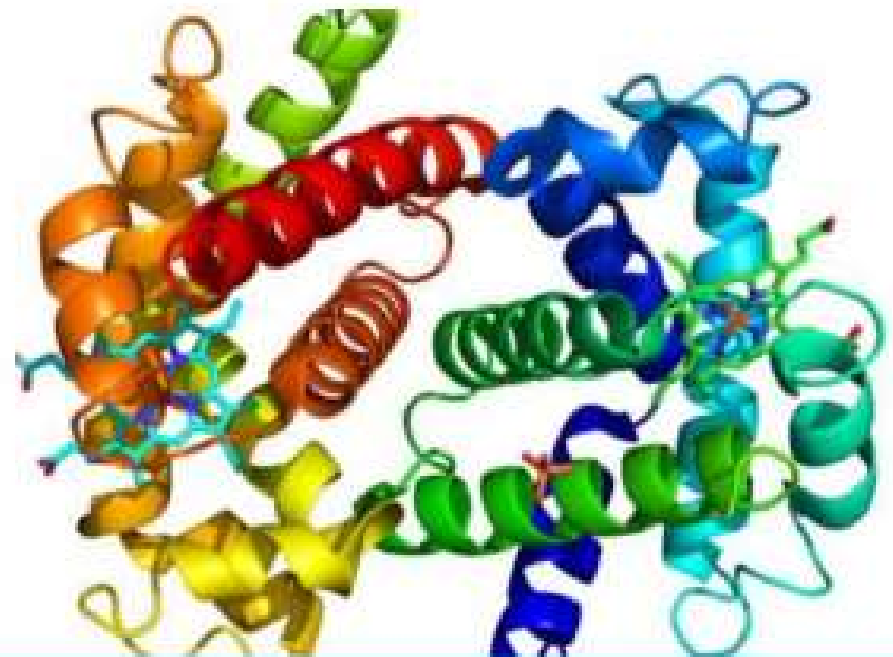


1.3 Protein:



Proteins are large biomolecules, or macromolecules, consisting of one or more long chains of **amino acid** residues.

Polypeptides





2. Chemical Structure



Chemical Structure of
Peptides & Proteins

2.1 Chemical Structure of Peptides



Number of Amino Acids

Dipeptides: Two amino acids with a single peptide bond.

Tripeptides: Three amino acids with two peptide bonds.

Tetrapeptides: Four amino acids with three peptide bonds.

And so on... ..

A **polypeptide** is a long, continuous, and unbranched peptide chain.

2.2 Chemical Structure of Proteins



Four Levels of Protein Structures

1. Primary Structure

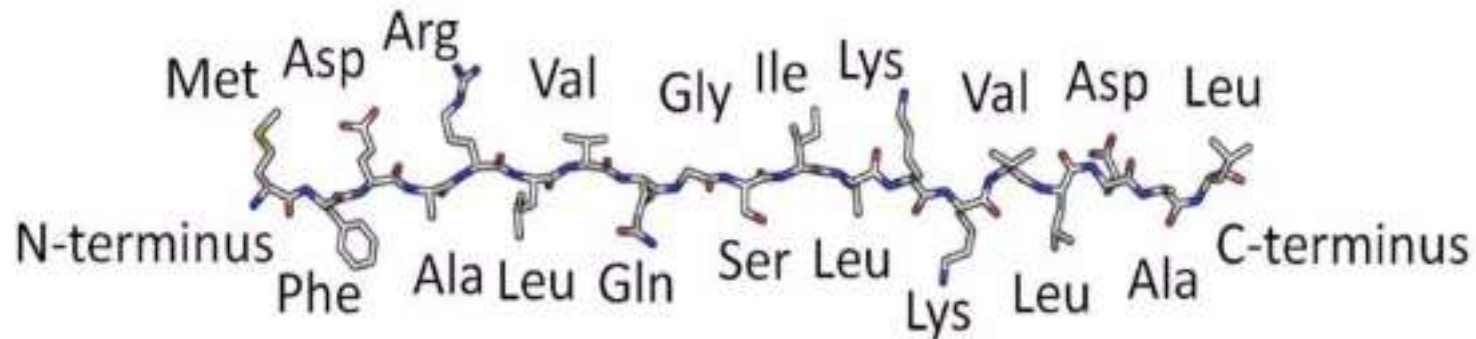
2. Secondary Structure

3. Tertiary Structure

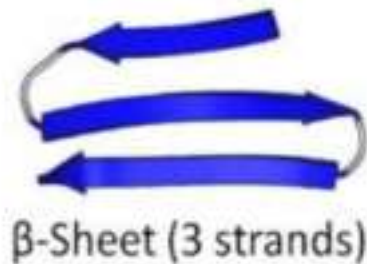
4. Quaternary Structure

Chemical Structure of Proteins

1. **Primary Structure:** The exact specification of its atomic composition and the chemical bonds connecting those atoms.



2. **Secondary Structure:** The three dimensional form of *local segments* of proteins. The two most common secondary structural elements are **α -helix** and **β -sheet**.

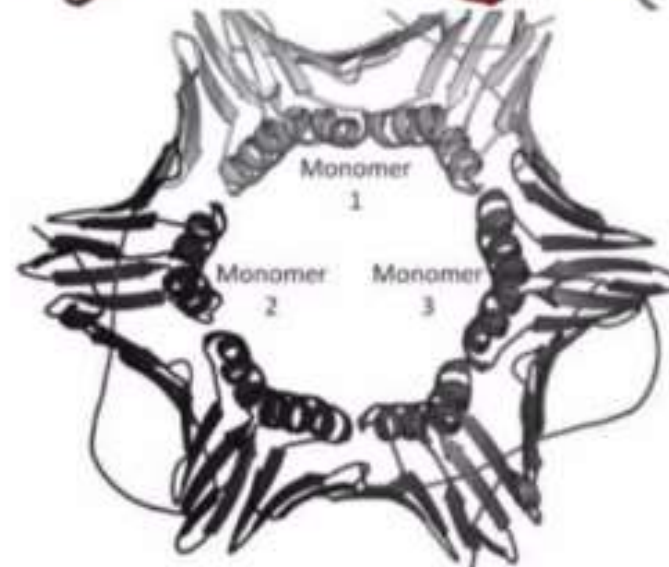


2.2 Chemical Structure of Proteins

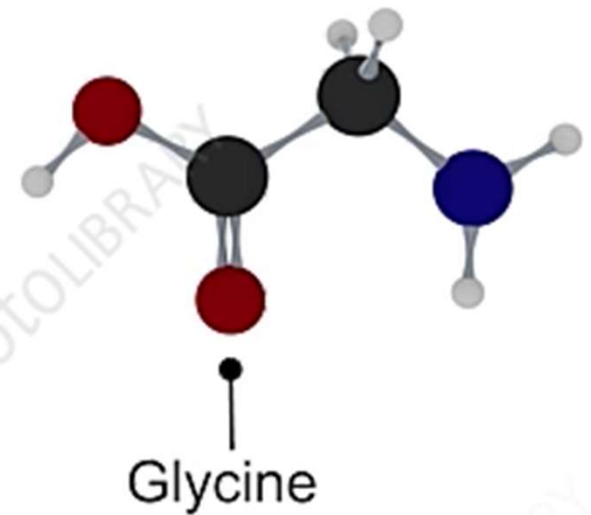
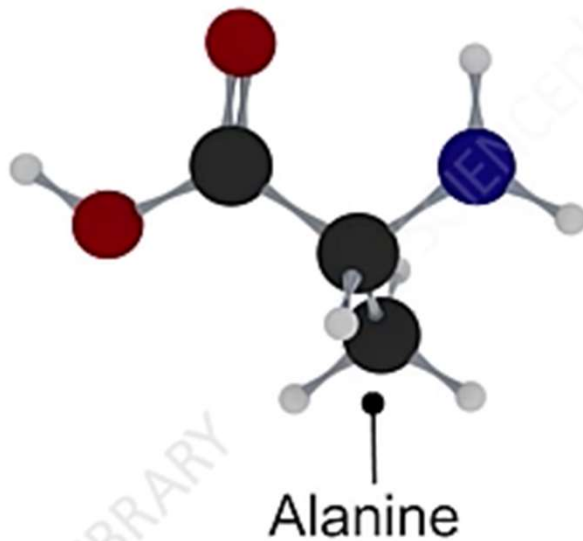
3. **Tertiary Structure:** The three dimensional shape of a protein. The tertiary structure will have a single polypeptide chain "backbone" with one or more protein secondary structures, the protein domains.



2. **Quaternary Structure:** The structure formed by several protein molecules (polypeptide chains), usually called *protein subunits* in this context, which function as a single protein complex.



Structure of Proteins



3. Classifications

Classifications of
Peptides & Proteins

3.1. Classifications of Peptides

Milk peptides:

Formed from the milk protein casein when digestive enzymes break this down.

Ribosomal peptides:

Synthesized by translation of mRNA.

Non-ribosomal peptides:

Assembled by enzymes that are specific to each peptide, rather than by the ribosome.

3.2. Classifications of Proteins

Globular Proteins:

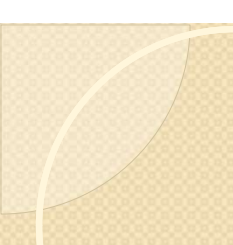
Almost all globular proteins are soluble and many are enzymes.

Fibrous Proteins :

These are often structural, such as collagen, the major component of connective tissue, or keratin, the protein component of hair and nails.

Membrane proteins :

They often serve as receptors or provide channels for polar or charged molecules to pass through the cell membrane.



4. Metabolism

Protein Metabolism:

Protein metabolism denotes the various biochemical processes responsible for the synthesis of proteins and amino acids, and the breakdown of proteins (and other large molecules, too) by catabolism.

- **Protein catabolism** is the process by which proteins are broken down to their amino acids.
This is also called **Proteolysis**.

5. Synthesis

The process of production
of Peptides and Proteins

5.1 Peptide Synthesis

5.2 Protein Synthesis

5.1. Peptide Synthesis

The process in which Peptides are built.

Peptides are synthesized by **coupling** the **carboxyl group** of one amino acid to the amino group of another amino acid molecule. Due to the possibility of **unintended reactions**, protecting groups are usually necessary. Chemical peptide synthesis most commonly starts at the **carboxyl** end of the **peptide**, and proceeds toward the **amino-terminus**.

5.2. Protein Synthesis

The process in which cells build Proteins.
(Also known as Protein Biosynthesis)

The term
is sometimes used to refer only to
protein translation but more often
it refers to a multi-step process,
beginning with amino acid synthesis
and **transcription** of nuclear **DNA**
into messenger **RNA**, which is then
used as input to **translation**.

6. Peptides in study



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- Peptides have recently been used in the study of **protein structure and function**.
 - For example, **synthetic peptides** can be used as **probes** to see where **protein-peptide interactions** occur.
 - Inhibitory peptides** are also used in clinical research to examine the effects of peptides on the inhibition of **cancer proteins** and other diseases.

7. Functions of Proteins

Antibodies

are specialized proteins involved in defending the body from antigens (foreign invaders). One way antibodies destroy antigens is by immobilizing them so that they can be destroyed by white blood cells.

Contractile Proteins

are responsible for movement. Examples include actin and myosin. These proteins are involved in muscle contraction and movement.

7. Functions of Proteins

Enzymes

are proteins that facilitate biochemical reactions. They are often referred to as catalysts because they speed up chemical reactions. Examples include the enzymes lactase and pepsin. Lactase breaks down the sugar lactose found in milk. Pepsin is a digestive enzyme that works in the stomach to break down proteins in food.

Transport Proteins

are carrier proteins which move molecules from one place to another around the body. Examples include hemoglobin and cytochromes. Hemoglobin transports oxygen through the blood. Cytochromes operate in the electron transport chain as electron carrier proteins.

7. Functions of Proteins

Hormonal Proteins

are messenger proteins which help to coordinate certain bodily activities. Examples include insulin, oxytocin, and somatotropin. Insulin regulates glucose metabolism by controlling the blood-sugar concentration. Oxytocin stimulates contractions in females during childbirth. Somatotropin is a growth hormone that stimulates protein

Transport Proteins

are fibrous and stringy and provide support. Examples include

keratin, collagen, and elastin. Keratins strengthen protective coverings such as hair, quills, feathers, horns, and beaks. Collagens and elastin provide support for connective tissues such as tendons and ligaments.

